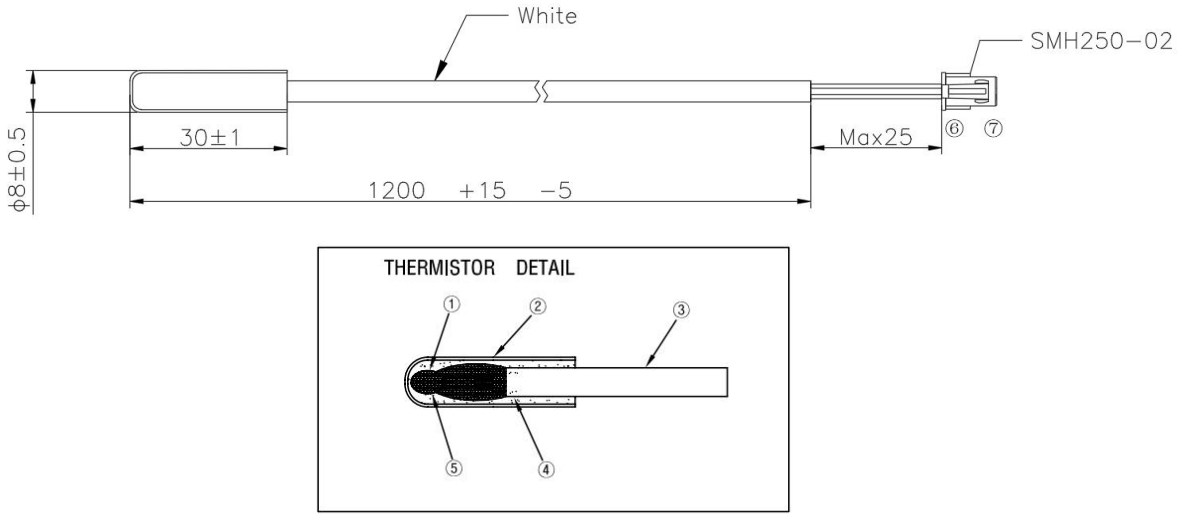
Product Specification	PART NO	CWF-DT2-103F3977FB1200G11A-SMH250
	FILE NO	QCZH-XS-20220222-A

NTC Temperature sensor

1、外形尺寸 Shape Dimensions

（单位 Unit：毫米mm）



2、产品组成 Component

序号 Part	材料名称 Definition	规格 Specification	数 量 Quantity
1.	热敏电阻 NTC	**R25**=10K±1%；**B25/85**=3977±1%	1PCS
2.	电线 Wire	PVC 护套线 22AWG 105℃ White 1200mm	1PCS
3.	传感器探头 Sensor probe	PP 材质 OD8*30mm	1PCS
4.	灌封 Encapsulation	环氧树脂 Epoxy	--
5.	连接端子 Plastic terminal	YST025 SMH250-02 White	2 1

3、电气性能 Electric Feature

序号 Part	项目 Items	测试条件 Measure methods	最小值 Min	正常值 Nominal	最大值 Max	单位 Unit
3-1.	25℃电阻值(R25) Resistance at 25℃	Ta=25±0.05℃ PT≦0.1mw	9.9	10	10.1	KΩ
3-2.	B 值($B_{25/85}$) B Value	$B=LN\frac{R_{T1}}{R_{T2}}/(\frac{1}{T1}-\frac{1}{T2})$	3937.23	3977	4016.77	K
3-3.	耗散系数(σ) Dissipation	Ta=25±0.5℃	0.7	/	/	mw/℃

3-4.	时间常数(τ) Thermal Time Constant	25℃空气到85℃水中,电阻变化63.2% 25℃ air to 85℃ stirred water	5	/	15	sec
3-5.	耐压测试 Hi-pot Test	1500VAC/2mA	/	5	/	S
3-6.	使用温度范围 Operating temperature range	/	-20	/	80	℃

4、产品命名方法 Naming Rule

CWF DT2 103 F 3977 F B 1200 B 11 A SMH250-02

① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨ ⑩ ⑪ ⑫

①	型号代码 Type Code	NTC 温度传感器 NTC Sensor
②	热敏电阻类型 Thermistor Type	DT,DT2,TA,TS,TJ,HT,CT
③	25℃电阻值 Resistance at 25℃	103＝10*10^3Ω＝10KΩ 104＝10*10^4Ω＝100KΩ
④	阻值精度 Resistance Tolerance	F=±1%,G=±2%,H=±3%,J=±5%
⑤	B值 B Value	3100,3270,3380,3435,3470,3950,4000,4150,4300,4500…
⑥	B值精度 B Value Tolerance	F=±1%,G=±2%,H=±3%,J=±5%
⑦	B值计算方式 B Value Calculating Methods	A：B25/50，B：B25/85,C：B0/25,D：B25/37.5,E：0/100,F：B100/200
⑧	线长 Wire Length	601=60*10^1mm，102＝10*10^2mm
⑨	封装类型/感温端子类型 Package Type/ Temperature terminal type	A-环氧医用型，B-表面测温型，C-子弹头型，D-环氧塑封型,E-法兰型，F-螺纹型 G-直管型，H-螺纹直管型，I-T 型，J-高温直管型,K-套管型，Z-组合型 A、Epoxy medical type, B、Surface temperature measurement type, C、Bullet type, D、Epoxy plastic sealing type, E、 Flange type, F、Thread type, G、Straight tube type, H、 Threaded straight tube type , I、T type, J、high temperature straight pipe type, K、casing type, Z、combined type
⑩	材质 Material	1、铜；2、黄铜；3、不锈钢；4、ABS；5、环氧树脂；6、PVC；7、TPE；8、玻璃纤维；9、薄膜 10、铝 11、PP 1.Copper，2.Brass，3.Stainless steel，4. ABS，5.Epoxy resin，6.PVC，7.TPE，8.Glass fiber，9.Film 10.Aluminum 11.PP
⑪	线材 Wire	A. PVC 线；B.TPE 线；C.XL-PE 线；D.硅胶线；E.Teflon 线；F.玻璃纤维线；G.漆包线；H.杜美丝；I.镀锡铜包钢线 A. PVC; B.TPE; C.XL-PE; D.Silicone wire; E.Teflon; F.Glass fiber wire; G.Enameled wire; H.Dumet; I.Tinned copper clad steel

⑫	连接端子类型 Connection Terminal Type	工厂内部代码 Factory Internal Code

5、机械性能 Mechanical Properties

序号 Part	项目 Items	技术要求 Criteria	测试条件及方法 Test Methods
5-1.	拉力测试 Tensile Test	树脂、电线粘接处不开裂,不变形、无损伤 No Crack Between Epoxy And Wire, No Deformation, No Break	加载 9.8N（1KGF）1 分钟 9.8N（1KGF）, 1 minute
5-2.	落地测试 Drop Test	无可见性损伤 No Visible damage	在 1 米的高度，让产品做自由落体运动，下落到 10mm 厚的橡木板上，5 次 Free Falling At The Height Of 1 Meter, 5 times

6、可靠性试验

序号 Part	项目 Items	技术要求 Criteria	测试条件及方法 Test Methods
6-1.	高温试验 Storage in dry heat	ΔR/R25≤±3% 无可见性损伤 No visible damage	105±5℃，1000±24h，DC0.2mA （参照 IEC60068-2-2/GB2423.2 试验）
6-2.	低温试验 Storage in coldness	ΔR/R25≤±3% 无可见性损伤 No visible damage	-30±5℃，1000±24h，DC0.2mA （参照 IEC60068-2-1/GB2423.1 试验）
6-3.	耐潮湿试验 Storage in damp, heat ,steady state	ΔR/R25≤±3% 无可见性损伤 No visible damage	40±2℃,90%-95%RH, 1000±24h （参照 IEC60068-2-3/GB2423.3 试验）
6-4.	温度冷热循环试验 Thermal shock	ΔR/R25≤±3% 无可见性损伤 No visible damage	-30℃×30min→25℃×5min→105℃×30min→25℃×5min，反复 5 次 5 times （参照 IEC60068-2-14/GB2423.22 试验）

7、产品使用条件(Product use conditions)

1)产品使用的最大工作温度，最大功率等，均依照规格书要求作业，不可超出规格书之范围

(The maximum service temperature of product use, the maximum power to wait, all in accordance with the specification requirements homework, shall not exceed the scope of specification)

2)产品移动、安装必须轻拿轻放，不可用力拉动

(Product mobile, installation must be handled with care not pulling force)

3）外壳发生变形、氧化等现象时，不要使用，以免影响感温精度

(Cap occur deformation, oxidation etc. Phenomenon, do not use, so as not to affect the temperature accuracy)

4）产品外观发现变形、破损时，不可使用，以免影响电性能

(Product appearance found deformation, breakage, do not use, lest affect performance)

5）在操作温度范围内，应尽量避免过于激烈的温度变化

(Avoid from exceeding radical temperature change,which is beyond operation temperature range)

6）不可施加过度振动的压力

(Do not add excessive vibration shocking pressure)

7)避免过度拉伸,弯曲导线

(Avoid from excessive pulling and bending of the lead wire)

8)不要使用在有腐蚀性气体(CO2,NH3,Sox,Nox)超出指定的条件。不要使用在电解、盐、酸、碱性和有机溶剂超出指定的条件

(Do not use in corrosiveness gas atmosphere (CO2,NH3,Sox,Nox) beyond the designated condition.) Do not use at the place where the sensor touches the electrolytic,brine, acid,alkaline and organic solvent beyond the designated condition)

9)尽可能避免使用在有水、高湿及其它带腐蚀性的环境中

(As far as possible avoid using in water, high humidity and other have rot environment.)

10)通过负温度系数温度传感器的电流会引起组件自身发热而产生测量误差，因此需在使用前将此 因素考虑在内.

(Through the negative temperature coefficient temperature sensor current will cause element own fever and produce measurement error, therefore need before use this factor into consideration.)

11）在极限温度下，产品可短期承受过高或过低的温度冲击，但不可长期将产品置于极限温度下，以避免缩短产品的使用寿命.

(In extreme temperatures, the product can be short-term bear too high or too low temperature impact,but do not put product at extreme temperatures for a long time, in order to avoid shortening the service life of products.)

8、R-T 表 R-T table

R_{25}=10kΩ±1%

$B_{25/85}$=3977K±1%

T(℃)	Rmin(kΩ)	Rcent(kΩ)	Rmax(kΩ)	DR(%)	DT(℃)	T(℃)	Rmin(kΩ)	Rcent(kΩ)	Rmax(kΩ)	DR(%)	DT(℃)
-40	262.38	273.95	286.00	4.4%	0.76	21	11.793	11.934	12.074	1.2%	0.26
-39	248.02	258.81	270.04	4.3%	0.76	22	11.285	11.414	11.543	1.1%	0.25
-38	234.46	244.53	254.99	4.3%	0.75	23	10.800	10.919	11.038	1.1%	0.24
-37	221.68	231.06	240.82	4.2%	0.74	24	10.339	10.448	10.557	1.0%	0.24
-36	209.62	218.37	227.46	4.2%	0.73	25	9.900	10.000	10.100	1.0%	0.23
-35	198.24	206.40	214.87	4.1%	0.72	26	9.473	9.573	9.673	1.0%	0.24
-34	187.51	195.12	203.01	4.0%	0.71	27	9.067	9.167	9.267	1.1%	0.25
-33	177.38	184.48	191.83	4.0%	0.70	28	8.681	8.780	8.879	1.1%	0.27
-32	167.83	174.44	181.29	3.9%	0.70	29	8.312	8.411	8.510	1.2%	0.28
-31	158.81	164.98	171.36	3.9%	0.69	30	7.962	8.059	8.158	1.2%	0.29
-30	150.30	156.05	162.00	3.8%	0.68	31	7.627	7.724	7.822	1.3%	0.30
-29	142.27	147.63	153.17	3.8%	0.67	32	7.309	7.405	7.502	1.3%	0.31
-28	134.69	139.69	144.85	3.7%	0.66	33	7.005	7.100	7.196	1.3%	0.32
-27	127.53	132.19	137.01	3.6%	0.65	34	6.716	6.810	6.905	1.4%	0.34
-26	120.78	125.12	129.61	3.6%	0.65	35	6.440	6.533	6.626	1.4%	0.35
-25	114.40	118.45	122.63	3.5%	0.64	36	6.177	6.268	6.361	1.5%	0.36

-24	108.38	112.15	116.05	3.5%	0.63	37	5.926	6.016	6.107	1.5%	0.37
-23	102.69	106.21	109.84	3.4%	0.62	38	5.686	5.775	5.865	1.6%	0.39
-22	97.324	100.60	103.98	3.4%	0.61	39	5.457	5.545	5.634	1.6%	0.40
-21	92.251	95.308	98.456	3.3%	0.61	40	5.239	5.326	5.413	1.6%	0.41
-20	87.459	90.309	93.241	3.2%	0.60	41	5.031	5.116	5.202	1.7%	0.42
-19	82.932	85.588	88.320	3.2%	0.59	42	4.832	4.915	5.000	1.7%	0.44
-18	78.654	81.130	83.674	3.1%	0.58	43	4.642	4.724	4.807	1.8%	0.45
-17	74.612	76.919	79.289	3.1%	0.57	44	4.460	4.541	4.622	1.8%	0.46
-16	70.791	72.941	75.149	3.0%	0.57	45	4.286	4.366	4.446	1.8%	0.47
-15	67.179	69.183	71.239	3.0%	0.56	46	4.120	4.198	4.277	1.9%	0.49
-14	63.764	65.632	67.547	2.9%	0.55	47	3.962	4.038	4.116	1.9%	0.50
-13	60.536	62.276	64.059	2.9%	0.54	48	3.810	3.885	3.961	2.0%	0.51
-12	57.482	59.104	60.765	2.8%	0.53	49	3.665	3.738	3.813	2.0%	0.53
-11	54.594	56.105	57.652	2.8%	0.52	50	3.526	3.598	3.672	2.0%	0.54
-10	51.862	53.269	54.710	2.7%	0.52	51	3.393	3.464	3.536	2.1%	0.55
-9	49.277	50.588	51.929	2.7%	0.51	52	3.266	3.335	3.406	2.1%	0.57
-8	46.831	48.052	49.300	2.6%	0.50	53	3.144	3.212	3.282	2.2%	0.58
-7	44.515	45.653	46.815	2.5%	0.49	54	3.028	3.094	3.162	2.2%	0.59
-6	42.324	43.383	44.465	2.5%	0.48	55	2.916	2.982	3.048	2.2%	0.61
-5	40.249	41.236	42.242	2.4%	0.48	56	2.809	2.873	2.938	2.3%	0.62
-4	38.284	39.203	40.139	2.4%	0.47	57	2.707	2.770	2.833	2.3%	0.63
-3	36.423	37.278	38.150	2.3%	0.46	58	2.609	2.670	2.733	2.3%	0.65
-2	34.660	35.457	36.267	2.3%	0.45	59	2.515	2.575	2.636	2.4%	0.66
-1	32.990	33.731	34.485	2.2%	0.44	60	2.425	2.483	2.543	2.4%	0.68
0	31.408	32.097	32.798	2.2%	0.44	61	2.338	2.396	2.454	2.5%	0.69
1	29.908	30.549	31.201	2.1%	0.43	62	2.255	2.312	2.369	2.5%	0.71
2	28.486	29.082	29.689	2.1%	0.42	63	2.176	2.231	2.287	2.5%	0.72
3	27.138	27.693	28.256	2.0%	0.41	64	2.099	2.153	2.208	2.6%	0.73
4	25.859	26.375	26.898	2.0%	0.40	65	2.026	2.079	2.133	2.6%	0.75
5	24.647	25.126	25.612	1.9%	0.40	66	1.956	2.007	2.060	2.6%	0.76
6	23.496	23.942	24.393	1.9%	0.39	67	1.888	1.939	1.991	2.7%	0.78
7	22.405	22.818	23.238	1.8%	0.38	68	1.823	1.873	1.924	2.7%	0.79
8	21.369	21.753	22.142	1.8%	0.37	69	1.761	1.810	1.859	2.7%	0.81
9	20.385	20.742	21.103	1.7%	0.36	70	1.701	1.749	1.797	2.8%	0.82
10	19.452	19.783	20.117	1.7%	0.35	71	1.644	1.690	1.738	2.8%	0.84
11	18.565	18.872	19.182	1.6%	0.35	72	1.589	1.634	1.681	2.8%	0.85
12	17.723	18.008	18.295	1.6%	0.34	73	1.536	1.580	1.626	2.9%	0.87
13	16.923	17.187	17.453	1.5%	0.33	74	1.485	1.528	1.573	2.9%	0.88
14	16.163	16.407	16.654	1.5%	0.32	75	1.436	1.478	1.522	2.9%	0.90
15	15.441	15.667	15.895	1.5%	0.31	76	1.388	1.430	1.473	3.0%	0.91
16	14.754	14.963	15.174	1.4%	0.30	77	1.343	1.384	1.425	3.0%	0.93
17	14.101	14.295	14.489	1.4%	0.30	78	1.299	1.339	1.380	3.1%	0.94
18	13.480	13.659	13.839	1.3%	0.29	79	1.257	1.296	1.336	3.1%	0.96
19	12.890	13.055	13.221	1.3%	0.28	80	1.217	1.255	1.294	3.1%	0.98
20	12.328	12.480	12.633	1.2%	0.27						

T(℃)	Rmin(kΩ)	Rcent(kΩ)	Rmax(kΩ)	DR(%)	DT(℃)	T(℃)	Rmin(kΩ)	Rcent(kΩ)	Rmax(kΩ)	DR(%)	DT(℃)
81	1.178	1.215	1.254	3.2%	0.99	141	0.2223	0.2331	0.2445	4.9%	2.09
82	1.141	1.177	1.214	3.2%	1.01	142	0.2171	0.2277	0.2389	4.9%	2.11
83	1.104	1.140	1.177	3.2%	1.02	143	0.2120	0.2225	0.2334	4.9%	2.13
84	1.070	1.105	1.140	3.2%	1.04	144	0.2071	0.2174	0.2281	4.9%	2.16
85	1.036	1.070	1.105	3.3%	1.06	145	0.2023	0.2124	0.2230	5.0%	2.18
86	1.004	1.037	1.072	3.3%	1.07	146	0.1977	0.2076	0.2179	5.0%	2.20
87	0.9728	1.005	1.039	3.3%	1.09	147	0.1932	0.2029	0.2131	5.0%	2.22
88	0.9428	0.9748	1.008	3.4%	1.10	148	0.1888	0.1983	0.2083	5.0%	2.24
89	0.9139	0.9452	0.9775	3.4%	1.12	149	0.1845	0.1939	0.2037	5.1%	2.26
90	0.8861	0.9167	0.9482	3.4%	1.14	150	0.1804	0.1896	0.1992	5.1%	2.29
91	0.8592	0.8891	0.9200	3.5%	1.15	151	0.1763	0.1854	0.1948	5.1%	2.31
92	0.8333	0.8626	0.8928	3.5%	1.17	152	0.1724	0.1813	0.1906	5.1%	2.33
93	0.8083	0.8370	0.8666	3.5%	1.19	153	0.1686	0.1773	0.1864	5.2%	2.35
94	0.7842	0.8122	0.8412	3.6%	1.20	154	0.1649	0.1734	0.1824	5.2%	2.38
95	0.7609	0.7883	0.8167	3.6%	1.22	155	0.1613	0.1697	0.1785	5.2%	2.40
96	0.7384	0.7653	0.7930	3.6%	1.24	156	0.1577	0.1660	0.1747	5.2%	2.42
97	0.7167	0.7430	0.7702	3.7%	1.25	157	0.1543	0.1624	0.1710	5.2%	2.44
98	0.6957	0.7215	0.7481	3.7%	1.27	158	0.1510	0.1590	0.1673	5.3%	2.47
99	0.6755	0.7007	0.7268	3.7%	1.29	159	0.1477	0.1556	0.1638	5.3%	2.49
100	0.6560	0.6806	0.7062	3.8%	1.31	160	0.1446	0.1523	0.1604	5.3%	2.51
101	0.6371	0.6612	0.6862	3.8%	1.32	161	0.1415	0.1491	0.1570	5.3%	2.54
102	0.6188	0.6425	0.6670	3.8%	1.34	162	0.1385	0.1459	0.1538	5.4%	2.56
103	0.6012	0.6244	0.6483	3.8%	1.36	163	0.1356	0.1429	0.1506	5.4%	2.59
104	0.5842	0.6069	0.6303	3.9%	1.38	164	0.1327	0.1399	0.1475	5.4%	2.61
105	0.5677	0.5899	0.6129	3.9%	1.39	165	0.1300	0.1370	0.1445	5.4%	2.63
106	0.5518	0.5735	0.5961	3.9%	1.41	166	0.1273	0.1342	0.1415	5.4%	2.66
107	0.5364	0.5577	0.5797	4.0%	1.43	167	0.1247	0.1315	0.1387	5.5%	2.68
108	0.5215	0.5423	0.5640	4.0%	1.45	168	0.1221	0.1288	0.1359	5.5%	2.71
109	0.5071	0.5275	0.5487	4.0%	1.47	169	0.1196	0.1262	0.1332	5.5%	2.73
110	0.4932	0.5132	0.5339	4.0%	1.48	170	0.1172	0.1237	0.1305	5.5%	2.76
111	0.4797	0.4993	0.5196	4.1%	1.50	171	0.1148	0.1212	0.1279	5.6%	2.78
112	0.4666	0.4858	0.5057	4.1%	1.52	172	0.1125	0.1188	0.1254	5.6%	2.81
113	0.4540	0.4728	0.4923	4.1%	1.54	173	0.1102	0.1164	0.1229	5.6%	2.83
114	0.4418	0.4602	0.4793	4.2%	1.56	174	0.1080	0.1141	0.1205	5.6%	2.86
115	0.4299	0.4480	0.4667	4.2%	1.58	175	0.1059	0.1119	0.1182	5.6%	2.89
116	0.4185	0.4362	0.4545	4.2%	1.60	176	0.1038	0.1097	0.1159	5.7%	2.91
117	0.4074	0.4247	0.4427	4.2%	1.61	177	0.1018	0.1076	0.1137	5.7%	2.94
118	0.3966	0.4136	0.4313	4.3%	1.63	178	0.0998	0.1055	0.1115	5.7%	2.96
119	0.3862	0.4029	0.4202	4.3%	1.65	179	0.0979	0.1035	0.1094	5.7%	2.99
120	0.3761	0.3924	0.4094	4.3%	1.67	180	0.0960	0.1015	0.1073	5.7%	3.02
121	0.3664	0.3823	0.3990	4.4%	1.69	181	0.0941	0.0996	0.1053	5.8%	3.04
122	0.3569	0.3725	0.3889	4.4%	1.71	182	0.0923	0.0977	0.1033	5.8%	3.07
123	0.3477	0.3631	0.3790	4.4%	1.73	183	0.0906	0.0959	0.1014	5.8%	3.10
124	0.3388	0.3538	0.3695	4.4%	1.75	184	0.0889	0.0941	0.0995	5.8%	3.13

125	0.3302	0.3449	0.3603	4.5%	1.77	185	0.0872	0.0923	0.0977	5.8%	3.15
126	0.3218	0.3363	0.3513	4.5%	1.79	186	0.0856	0.0906	0.0959	5.9%	3.18
127	0.3137	0.3279	0.3427	4.5%	1.81	187	0.0840	0.0889	0.0942	5.9%	3.21
128	0.3058	0.3197	0.3342	4.5%	1.83	188	0.0824	0.0873	0.0925	5.9%	3.24
129	0.2982	0.3118	0.3260	4.6%	1.85	189	0.0809	0.0857	0.0908	5.9%	3.27
130	0.2908	0.3041	0.3181	4.6%	1.87	190	0.0794	0.0842	0.0892	5.9%	3.29
131	0.2836	0.2967	0.3104	4.6%	1.89	191	0.0780	0.0827	0.0876	6.0%	3.32
132	0.2766	0.2895	0.3029	4.6%	1.91	192	0.0766	0.0812	0.0860	6.0%	3.35
133	0.2698	0.2825	0.2956	4.7%	1.93	193	0.0752	0.0797	0.0845	6.0%	3.38
134	0.2633	0.2756	0.2886	4.7%	1.95	194	0.0739	0.0783	0.0830	6.0%	3.41
135	0.2569	0.2690	0.2817	4.7%	1.97	195	0.0726	0.0769	0.0816	6.0%	3.44
136	0.2507	0.2626	0.2751	4.7%	1.99	196	0.0713	0.0756	0.0802	6.1%	3.47
137	0.2447	0.2564	0.2686	4.8%	2.01	197	0.0700	0.0743	0.0788	6.1%	3.50
138	0.2388	0.2503	0.2623	4.8%	2.03	198	0.0688	0.0730	0.0774	6.1%	3.53
139	0.2331	0.2444	0.2562	4.8%	2.05	199	0.0676	0.0717	0.0761	6.1%	3.56
140	0.2276	0.2387	0.2502	4.8%	2.07	200	0.0664	0.0705	0.0748	6.1%	3.59